Operation, strengths and limitations of solar power as a method of electricity generation to meet both energy needs and climate change goals.

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University of Birmingham Assessment and Feedback Student Template

Section 1:

- I have gotten feedback that my referencing wasn't done in the correct way.
- I have also gotten feedback that the language used was not very formal and rather too much first-person language.
- I received feedback about including more visuals like pictures and tables, which was considered less critical.

Section 2:

- One of the actions I took is that I started writing references in this way – [], as well as using the DIO link and not the URL, this improved the flow of my report.
- Another improvement I have made is going from first-person language to formal language and speaking in a more general way.
- I easily added pictures, tables and diagrams with proven statistics, this also made my report more enjoyable and easier to read and understand.

Section 3:

- I would like to request feedback on the content.
- I would like to also request feedback on the validity of the content from linked evidence and references and how structured and organized the report is, if it goes in a nice flow or not.
- I would also like feedback on the clarity and formality of the language used.

Report Summary

Solar power, a key renewable energy source, is harnessed through solar thermal plants and solar cells. Solar thermal plants concentrate sunlight with mirrors to heat fluids like water, powering turbines, while silicon solar cells use semiconductor materials to generate direct current, converted to alternating current for grid use via an inverter.

Combining solar power with systems like Vapor Compression Refrigeration (VCRS) and Thermoelectric Cooling (TEC) enables efficient off-grid cooling, valuable in areas with limited electricity access. Despite these benefits, solar power has limitations. Concentrated solar power plants (CSPPs) lack operational flexibility and are slow to start due to mechanical constraints. Moreover, pollution and poor weather reduce solar effectiveness.

For regions like Small Island Developing States (SIDS), solar adoption strengthens energy security and cuts greenhouse gas emissions, supporting climate resilience and sustainable development amid the challenges of climate change.

Solar Power

The What and How of Solar Power

Solar power is one of the most important sources of energy, the sun can be used to generate electricity in two ways, either by using its heat as a heat source or by utilizing its light in a solar cell [1]. Solar thermal power plants use the sun as a heat source, to achieve enough temperature for a power plant, solar energy must be concentrated so mirrors are put into it, these power plants use solar heat to heat a thermodynamic fluid such as water in order to drive a thermodynamic engine; for water this will be a steam turbine [2].Silicon solar cells are connected in series and parallel, they are then enclosed in protective material. The cell output is normally direct current but must be converted into alternating current at the grid frequency so that the solar installation can connect to the grid, this conversion is carried out by an inverter [3]. The solar cells exploit properties inherent in semiconducting materials to capture photons of light and use them to generate an electric current [4].

Advantages Of Solar Power

The combination of refrigeration systems and solar photovoltaic (PV) technology has become a viable alternative to tackle the difficulties caused by electricity limitations, especially in areas with restricted grid connectivity. Among the various systems, Vapor Compression Refrigeration Systems (VCRS) and Thermoelectric Cooling (TEC) are widely known and easily accessible. Compression refrigeration powered by photovoltaic (PV) systems has become popular due to its high cooling efficiency (Yumrutaş et al.). It absorbs heat during the day and releases it at night, making it the most efficient cooling technique currently available (R. M. et al., 2019).

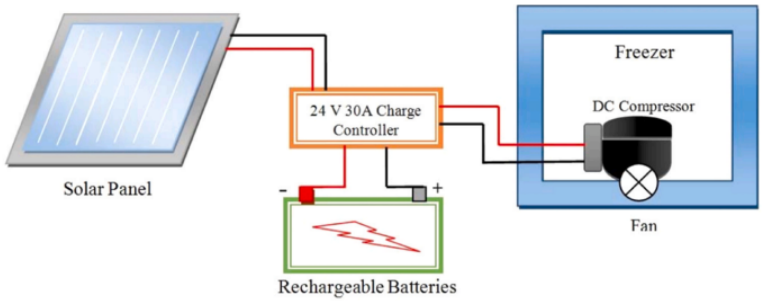


Fig. 2. Schematic diagram of a system with DC compressor (Iqbal et al., 2018).

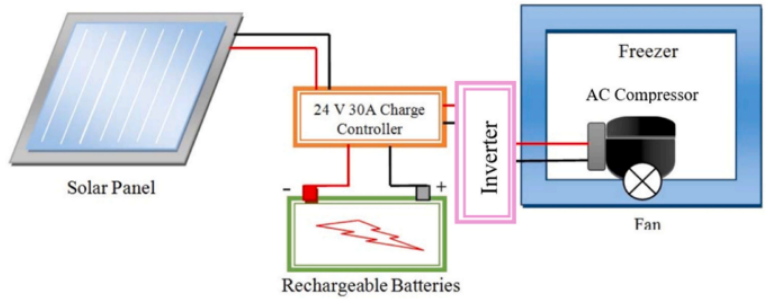


Fig. 3. Schematic diagram of a system with AC compressor (Iqbal et al., 2018).

For Fig 2: An electric motor transforms electrical energy into mechanical energy within a refrigeration compressor. An AC compressor is operated using alternating voltage, often at either 120 or 220 V and a frequency of 50 Hz.

For Fig 3: The system Solar's panel is managed by a charge controller which stabilises the voltage for compressor and batteries, DC compressors operate at low voltage of 12-24V with brushless motor unlike AC compressors, they do not need a power inverter [5].

Limitations of Solar Power

Concentrated solar power plants (CSPPs) are not currently designed for continuous operation, therefore they still experience daily start-ups and shut-downs. To maximize their performance from both technical and economical standpoints, increasing the flexibility of their dynamic performance is an important aspect which must be addressed (Topel et al., 2017). The rate at which a power plant can start up is limited by thermo-mechanical constraints, which may increase the time to reach the nominal load of the power plant. The receiver, the steam turbine and the steam generator system (SGS) are usually the most limiting components in this regard. While the receiver (Samanes and Garcia-Barberena, 2014) and steam turbine (Topel et al., 2017) have been examined in the literature, the steam generator has not been the focus of many studies [6]

Another massive issue with solar power is that it is unreliable, its effectiveness is compromised by atmospheric and meteorological conditions, particularly air pollution, which reduces solar radiation and panel efficiency.

This study was done in South Korea and using hourly power generation data from 2006 to 2013 and addressing potential endogeneity of PM10 with an instrumental variable approach, we find that a 10 mg/m^3 increase in PM10 reduces solar power generation by 2.17 MWh, resulting in an estimated annual economic loss of approximately USD 2.2 million during the study period [7]. The baseline model is represented by:

$$Y_t = \alpha + \beta_1 PM10_t + \beta_2 SD_t + \beta_3 X_t + \beta_4 f(t) + \varepsilon_t$$

The study analyses the impact of various factors on solar power generation (**Yt**) at a given time **t**. These factors include:

PM10t: Concentration of particulate matter smaller than 10 µm

SDt: Sunshine duration

Xt: Other meteorological conditions like precipitation, temperature, wind speed, and cloud cover

To account for potential non-linear trends, a time trend function (**f(t)**) with linear and quadratic terms is used. This model helps capture how air quality and weather variables affect solar energy output over time [7].

Contribution to Energy Needs and Climate Change

Facing rising climate change impacts, 39 Small Island Developing States (SIDS) are leading the global push for sustainable, resilient energy solutions.

Table 1 lists all fossil carbon dioxide (CO2) emissions in five of these countries. Table 1:

Country name	Per capita (t/cap)	Per country (kt)	Per energy unit (kg/kWh)
Asia-Pacific			
Cook Islands	4.0	68	0.14
Fiji	1.2	1,000	0.13
Kiribati	0.5	68	0.22

Tables 3 list installed solar energy capacity:

Table 3. Installed solar energy capacity in SIDS (in MW) in 2014–2023

Country	2014	2017	2020	2023
Asia-Pacific				
Cook Islands	2	3	6	6
Fiji	2	8	8	11
Kiribati	1	3	3	3

By harnessing solar power, SIDS can and strengthen their energy security as shown in Table 3. Switching to renewable energy reduces greenhouse gas (GHG) emissions, thereby lowering their role in causing climate change, this has been shown in Table 3 as installation of solar energy occurred there, we are able to tell that carbon dioxide (CO2) emissions have now definitely decreased [8].

Appendix/ChatGPT section

This section was written with the best of my ability

1. I used chat GPT to summarize the whole assignment page so I know what to do in an easier way

5 advantages of chat GPT for (1):

. It did it quickly

. I didn't have to read and comprehend all pages so that saved me some time

. It now has an idea of what I am doing so it could help with further questions more

. It was able to explain to me in easier language than what was used

. It was able to let me know the main important details rather than everything

5 disadvantages of chat GPT for (1):

. It may have missed important parts as it summarized it

. It didn't give a very in-depth explanation of what was needed to do

. Since the language used was lower level, I couldn't understand exactly what was needed from me

. It didn't summarize it too much and I still had to read a lot

. It didn't explain the mark scheme too well

2. It help me learn and write things and summarize things:

The following prompts were input into [Chat GPT]: <meaning of These modules are subsequently encapsulated to protect them from natural elements before they are deployed>

The output obtained was: <This sentence means that after the solar cells are combined into larger units called modules (or panels), they are then enclosed in a protective material or casing. This encapsulation process safeguards the modules from environmental factors like rain, dust, and sunlight damage. Once protected, the modules are ready to be installed or put into use.>

The output was changed by me in the following ways: <they are then enclosed in protective material>

The following prompts were input into [Chat GPT]:<Among all the systems, VCRS and TEC are widely recognized and easily obtainable. However, there are certain limitations associated with the TEC system. Several studies have indicated that TEC is appropriate for cold storage purposes, but not for freezing (Xu et al., 2017). In their study, Dai et al. demonstrated that the system achieved a COP of 0.3 and maintained a refrigerator temperature inside the cabinet between 5–10 ◦C (Dai et al., 2003). PV-driven compression refrigeration has gained widespread use due to its high cooling coefficient (Yumrutas¸ et al., throughout the day and releases it during the night. It is currently regarded as the most

efficient cooling method available (R. M. et al., 2019). Integrating PCM can improve the efficiency of PV panels, resulting in increased output.>

The output obtained was: <Among the various systems, Vapor Compression Refrigeration Systems (VCRS) and Thermoelectric Cooling (TEC) are widely known and easily accessible. However, the TEC system comes with certain limitations. Several studies have shown that TEC is suitable for cold storage applications but not effective for freezing (Xu et al., 2017). In their research, Dai et al. reported that the TEC system achieved a coefficient of performance (COP) of 0.3 and kept the refrigerator's internal temperature between 5 and 10 °C (Dai et al., 2003). Compression refrigeration powered by photovoltaic (PV) systems has become popular due to its high cooling efficiency (Yumrutaş et al.). It absorbs heat during the day and releases it at night, making it the most efficient cooling technique currently available (R. M. et al., 2019). The integration of phase change materials (PCM) can further enhance the efficiency of PV panels, leading to higher energy output.>

The output was changed by me in the following ways:<Among the various systems, Vapor Compression Refrigeration Systems (VCRS) and Thermoelectric Cooling (TEC) are widely known and easily accessible. Compression refrigeration powered by photovoltaic (PV) systems has become popular due to its high cooling efficiency (Yumrutaş et al.). It absorbs heat during the day and releases it at night, making it the most efficient cooling technique currently available (R. M. et al., 2019). >

The following prompts were input into [Chat GPT]: <summarize a lot The system uses a solar panel as a power generation source. The charge controller is responsible for delivering power to the compressor and batteries. The charge controller provides a consistent voltage to other components, compensating for the variable voltage of the solar panel caused by changing levels of irradiation. The DC compressor necessitates a low voltage and a direct current power source, often ranging from 12 to 24 volts. While DC compressors have a comparable mechanical compression component with AC compressors, they are distinguished by their brushless electrical motor. Green energy solutions have the capability to directly power a DC compressor, whereas an AC compressor necessitates the use of a power inverter, as illustrated in Fig. 3 (Ekren et al., 2013). >

The output obtained was: <The system uses a solar panel to generate power, which is managed by a charge controller. This controller stabilizes the voltage for the compressor and batteries, adjusting for fluctuations in sunlight. DC compressors operate on low voltage (12-24V) with a brushless motor, unlike AC compressors. They can be directly powered by green energy, while AC compressors need a power inverter (Ekren et al., 2013). >

The output was changed by me in the following ways: <The system solar’s panel is managed by a charge controller which stabilises the voltage for compressor and batteries, DC compressors operate at low voltage of 12-24V with brushless motor unlike AC compressors, they do not need a power inverter.>

The following prompts were input into [Chat GPT]: <meaning of Using hourly power generation data from 2006 to 2013 and addressing potential endogeneity of PM10 with an instrumental variable approach, we find that a 10 mg/m3 increase in PM10 reduces solar power generation by 2.17 MWh, resulting in an estimated annual economic loss of approximately USD 2.2 million during the study period.>

The output obtained was: <This statement explains a study that analysed the impact of air pollution (specifically PM10 particles) on solar power generation. The researchers used hourly power generation data from 2006 to 2013. They addressed the potential endogeneity (which refers to possible bias or correlation between variables) of PM10 levels by using an instrumental variable approach (a statistical method to correct for such biases). Their findings

indicate that for every 10 mg/m³ increase in PM10, solar power generation decreases by 2.17 megawatt-hours (MWh). This reduction in power output led to an estimated annual economic loss of around USD 2.2 million over the study period.>

The output was changed by me in the following ways: <This study was done in South Korea and using hourly power generation data from 2006 to 2013 and addressing potential endogeneity of PM10 with an instrumental variable approach, we find that a 10 mg/m^3 increase in PM10 reduces solar power generation by 2.17 MWh, resulting in an estimated annual economic loss of approximately USD 2.2 million during the study period.>

The following prompts were input into [Chat GPT]: <summarize a lot: where Yt represents solar power generation at time t, PM10t represents the concentration of particulate matter smaller than 10 μm in diameter at time t, SDt is sunshine duration at time t, and Xt includes other meteorological variables (precipitation, temperature, wind speed, and cloud cover) at time t. The function f(t) incorporates both linear and quadratic terms of the time trend variable t to control for potential nonlinear time trends over the research period.>

The output obtained was: <The study analyses the impact of various factors on solar power generation (**Yt**) at a given time **t**. These factors include:

- **PM10t**: Concentration of particulate matter smaller than 10 μm
- **SDt**: Sunshine duration
- **Xt**: Other meteorological conditions like precipitation, temperature, wind speed, and cloud cover

To account for potential non-linear trends, a time trend function (**f(t)**) with linear and quadratic terms is used. This model helps capture how air quality and weather variables affect solar energy output over time.>

The output was changed by me in the following ways: <The study analyses the impact of various factors on solar power generation (**Yt**) at a given time **t**. These factors include:

- **PM10t**: Concentration of particulate matter smaller than 10 μm
- **SDt**: Sunshine duration
- **Xt**: Other meteorological conditions like precipitation, temperature, wind speed, and cloud cover

To account for potential non-linear trends, a time trend function (**f(t)**) with linear and quadratic terms is used. This model helps capture how air quality and weather variables affect solar energy output over time.>

The following prompts were input into [Chat GPT]: <summarize In the wake of escalating climate change impacts, 39 Small Island Developing States (SIDS) find themselves at the forefront of the global call to action for sustainable and resilient energy solutions.>

The output obtained was: <Facing rising climate change impacts, 39 Small Island Developing States (SIDS) are leading the global push for sustainable, resilient energy solutions.

The output was changed by me in the following ways: Facing rising climate change impacts, 39 Small Island Developing States (SIDS) are leading the global push for sustainable, resilient energy solutions.>

The following prompts were input into [Chat GPT]: <summarize my whole 2 pages (I pasted the 2 pages)>

The output obtained was: <what I put into the report summary>

The output was changed by me in the following ways: <what I wrote in the report summary>

The following prompts were input into [Chat GPT]: <meaning of cuts greenhouse gas emissions, >

The output obtained was: <"Cuts greenhouse gas emissions" means reducing the release of gases like carbon dioxide (CO_2) and methane (CH_4) into the atmosphere. These gases trap heat and contribute to global warming, so by cutting or lowering these emissions, we lessen their impact on climate change.>

The output was changed by me in the following ways: <cuts greenhouse gas emissions>

5 advantages of using Chat GPT for (2):

. It gives fast replies

. It can summarize and keep all necessary information

. It can explain things very easily

. It was able to fit what I needed to write along with the mark scheme I provided it earlier I believe

. It allowed me to visualize situations will the in-depth explanations

5 disadvantages of using Chat GPT (2):

. Since it used easier language, it didn't give me a completely accurate explanation of the study

. It didn't summarize it too much and I still had to read a lot

. When summarizing, it sometimes neglects the topics I need summarized a lot and priorities the topics I need a little, both topics are in one sentence, but I need to write about one more.

. Sometimes it adds more unnecessary information when I ask it to summarize.

. When asking for a meaning of a sentence it sometimes tell me the meaning along with so much more unneeded information.

References

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